IIR Filter Design Procedure

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Key Concepts

1) Infinite impulse response (IIR) filters are usually designed by converting established continuous-time (analog) filter designs into discrete-time filters.

2) Continuous-time filter properties

    a) Frequency response $H(\Omega)$ measures the effect of the system on a complex sinusoid input $x(t) = e^{j\Omega t}$.

    b) The transfer function $H(s)$ is the continuous-time version of the discrete-time system function and measures the ef-

fect of the system on a damped complex sinusoid input $x(t) = e^{st} = e^{(\sigma+j\Omega)t} = e^{\sigma t}e^{j\Omega t}$

c) Linear time-invariant continuous-time systems can be described by linear constant coefficient differential equations. Such systems have rational transfer functions.

3) Five steps to IIR filter design (MATLAB performs steps b-d for you):

a) Select a prototype continuous-time lowpass filter.

b) Prewarping: convert the discrete-time critical frequencies ω_k (e.g., band edges) to continuous-time critical frequencies Ω_k.

c) Apply a frequency transformation to the prototype lowpass filter so that it satisfies the critical frequencies found in the previous step.

d) Use the bilinear transform to convert the continuous-time transfer function found in the previous step to a discrete-time system function.

e) Verify that your design satisfies the specifications.

Video Screenshots

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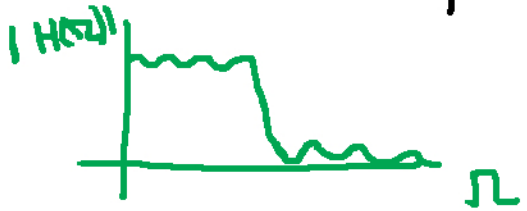
IIR discrete-time filters are designed from analog (continuous-time) prototypes

Butterworth - monotonic in both pass/stop bands

Chebyshev Type 1 - pass band equiripple, stop band monotonic

Chebyshev Type 2 - pass band monotonic, stop band equiripple

Elliptic - equiripple in both pass/stop bands



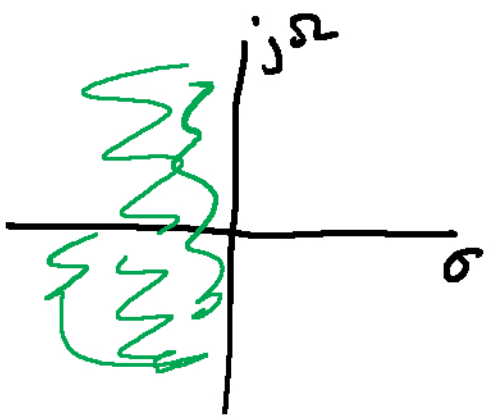
Continuous-time filters

Frequency response $H(\Omega)$ (Fourier transform)

Transfer function $H(s)$ (Laplace transform)

$$H(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt \qquad s = \sigma + j\Omega \quad \text{complex}$$

s-plane



$$H(\Omega) = H(s)\Big|_{\sigma=0}$$

Differential Equations

$$H(s) = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k} = \frac{b_M}{a_N} \frac{\prod_{k=1}^{M} (s - c_k)}{\prod_{k=1}^{N} (s - d_k)}$$

$$h(t) = \sum_{k=1}^{N} A_k e^{d_k t} u(t)$$

Stable/causal Systems

$\mathrm{Re}\{d_k\} < 0 \Rightarrow$ poles in left half of s-plane

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1) Prototype low pass filter $H_{LP}(s)$

2) Translate critical ω_k to Ω_k (prewarping)

3) Apply "frequency transformation" $s = f(\tilde{s})$

$$H(\tilde{s}) = H_{LP}(s)\Big|_{s=f(\tilde{s})} \Rightarrow H(\tilde{\Omega}) = H_{LP}(\Omega)\Big|_{\Omega=f(\tilde{\Omega})}$$

so critical frequencies of $H(\tilde{\Omega})$ are Ω_k

4) Use bilinear transform $\tilde{s} = 2\dfrac{1-z^{-1}}{1+z^{-1}}$

$$H(z) = H(\tilde{s})\Big|_{\tilde{s}=2\frac{1-z^{-1}}{1+z^{-1}}}$$

5) Verify that $H(z)$ satisfies the specifications

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